

HW1

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1 Ket notation

(a)It should satisfy that $|\psi\rangle$ is normalized:

$$\langle\psi|\psi\rangle = 1$$

$$\because \left|\frac{3}{C}\right|^2 + \left|\frac{-5i}{C}\right|^2 = 1$$

$$\therefore C = \sqrt{34}$$

$$|\psi\rangle = \frac{3}{\sqrt{34}}|0\rangle + \frac{-5i}{\sqrt{34}}|1\rangle$$

(b)

if two basis are orthonormal, they should satisfy that $\langle i | -i \rangle = 0$ and $\langle i | i \rangle = 1$ and $\langle -i | -i \rangle = 1$.

$$\because \langle i | -i \rangle = 0$$

$$\langle i | i \rangle = 1$$

$$\langle -i | -i \rangle = 1$$

$\therefore$ the basis is orthonormal

For the possible outcomes, we can calculate the probability of each outcome:

the probability of outcome $i = \langle i | \psi \rangle^2$

$$\begin{aligned} \langle i | \psi \rangle &= \frac{1}{\sqrt{34}} \cdot \frac{1}{\sqrt{2}} (\langle 0 | - i \langle 1 |) (3|0\rangle + -5i|1\rangle) \\ &= -\frac{2}{\sqrt{68}} \end{aligned}$$

$$\therefore P(i) = |\langle i | \psi \rangle|^2 = \frac{1}{17}$$

Calculating the probability of outcome $-i$ as well:

$$\begin{aligned} \langle -i | \psi \rangle &= \frac{1}{\sqrt{34}} \cdot \frac{1}{\sqrt{2}} (\langle 0 | + i \langle 1 |) (3|0\rangle + -5i|1\rangle) \\ &= \frac{8}{\sqrt{68}} \end{aligned}$$

$$\therefore P(-i) = |\langle -i | \psi \rangle|^2 = \frac{16}{17}$$

2 Unitary transformations.

(a)

For (i) $\rightarrow$ (ii), if we know that T is unitary, $T^*T = I$

$$\begin{aligned}\langle Tu, Tv \rangle &= (Tu)^*Tv \\ &= u^*T^*Tv \\ &= u^*Iv \\ &= \langle u, v \rangle\end{aligned}$$

For (ii) $\rightarrow$ (iii)

We know that

$$\begin{aligned}\|u\|^2 &= \langle u, u \rangle = 1 \\ \langle u, u \rangle &= \langle Tu, Tu \rangle \\ \|Tu\|^2 &= \|u\|^2 = 1\end{aligned}$$

Therefore, T maps unit vectors u to unit vectors Tu .

For (iii) $\rightarrow$ (i)

if a linear transformation T maps unit vectors to unit vectors, so for any unit vector u ,

$$\begin{aligned}\|Tu\|^2 &= \langle Tu, Tu \rangle \\ &= u^*T^*Tu \\ &= \langle u | T^*Tu \rangle \\ &= 1\end{aligned}$$

only when $T^*T = I$, this situation can be satisfied by any unit vector. Therefore, T is unitary.

(b)

To prove such linear transformation T exists.

We know T is a linear transformation, so we can write T as a matrix A .

$$\langle u, Tv \rangle = u^\dagger Av = (A^\dagger u)^\dagger v = \langle A^\dagger u, v \rangle$$

Therefore, if we define $T^\dagger$ as the linear transformation represented by $A^\dagger$, we can get that $\langle u, Tv \rangle = \langle T^\dagger u, v \rangle$.

To prove such linear transformation is unique, we can assume that there are two linear transformations T_1 and T_2 that satisfy the condition $\langle u, Tv \rangle = \langle T_1^\dagger u, v \rangle = \langle T_2^\dagger u, v \rangle$.

$$\begin{aligned}\therefore \langle T_1^\dagger u, v \rangle &= \langle T_2^\dagger u, v \rangle \\ \langle (T_1^\dagger - T_2^\dagger)u, v \rangle &= 0\end{aligned}$$

because for all u and $v \in \mathbb{C}^n$, it satisfied.

So, we know $T_1^\dagger = T_2^\dagger$.

Thus, we can conclude that such linear transformation exists and is unique.

(c)

(iii)

By the definition of unitary transformation, we know that for any $u, v \in \mathbb{C}^n$, $\langle Uu, Uv \rangle = \langle u, v \rangle$.

$$\begin{aligned}\langle Uu, Uv \rangle &= u^*U^*Uv = \langle u, U^*Uv \rangle \\ &= \langle u, v \rangle \therefore U^*U = UU^* = I\end{aligned}$$

(iii)

if we define the column of U as $u_1, u_2, \dots, u_n$, we can get that $U = [u_1, u_2, \dots, u_n]$.

from (ii), we can get that $U^*U = I$, so we can get that

$$\begin{bmatrix} u_1^*u_1 & u_1^*u_2 & \dots & u_1^*u_n \\ u_2^*u_1 & u_2^*u_2 & \dots & u_2^*u_n \\ \dots & \dots & \dots & \dots \\ u_n^*u_1 & u_n^*u_2 & \dots & u_n^*u_n \end{bmatrix} = I$$

Therefore, we can get that $u_i^*u_j = 0$ when $i \neq j$ and $u_i^*u_i = 1$ when $i = j$.

From the definition of

orthonormal basis (a set of vectors that are mutually orthogonal and each of them has unit length), we can conclude that the column of U forms an orthonormal basis of $\mathbb{C}^n$.

(iv)

if we define the row of U as $r_1, r_2, \dots, r_n$, we can get that $U = [r_1^T, r_2^T, \dots, r_n^T]^T$.

from (ii) we can get that $UU^* = I$, so we can get that $(UU^*)_{ij} = r_i r_j^*$,

So only when $i = j$, $r_i r_j^* = 1$, and when $i \neq j$, $r_i r_j^* = 0$.

From the definition of orthonormal basis (a set of vectors that are mutually orthogonal and each of them has unit length), we can conclude that the row of U forms an orthonormal basis of $\mathbb{C}^n$.

(d)

We know that $M(x, y) = \prod_{i=1}^n (-1)^{x_i y_i} = (-1)^{\sum_{i=1}^n x_i y_i} = (-1)^{x \cdot y}$.

So when x and y have the odd number of same 1s, $M(x, y) = -1$, and when x and y have the even number of same 1s, $M(x, y) = 1$.

we know for every column of M , it has 2^{n-1} or 1 , and if we want to make M correspond to a scalar multiple of unitary transformation, we set $M = cU$.

and $M = \{m_{\cdot i}, m_{\cdot 2}, \dots\}$, where m_i is the column of M .

we can calculate that $m_i^* m_i = 2^n$. so $|m_{\cdot i}| = \frac{1}{\sqrt{2^n}}$.

now we need to calculate $m_x^* m_y$ when $x \neq y$.

This time

$$\begin{aligned} m_y^* m_z &= \sum_{x \in \{0,1\}^n} M(x, y) M(x, z) \\ &= \sum_{x \in \{0,1\}^n} (-1)^{x \cdot y} (-1)^{x \cdot z} \\ &= \sum_{x \in \{0,1\}^n} (-1)^{x \cdot (y \oplus z)} \end{aligned}$$

because $y \neq z$, there is at least one bit that is different between y and z , so we substitute $y \oplus z$ with w and $w \neq 0$.

Then we can get

$$\begin{aligned} m_y^* m_z &= \sum_{x \in \{0,1\}^n} (-1)^{x \cdot w} \\ &= \sum_{x \in \{0,1\}^n} \prod_{i=1}^n (-1)^{x_i w_i} \\ &= \prod_{i=1}^n \sum_{x_i \in \{0,1\}} (-1)^{x_i w_i} \end{aligned}$$

because $w \neq 0$, there is at least one i such that $w_i = 1$, so $\sum_{x_i=\{0,1\}} (-1)^{x_i w_i} = 0$.
 Therefore, $m_y^* m_z = 0$ when $y \neq z$.

Thus, we can conclude that M corresponds to a scalar multiple of unitary transformation, and the scalar is $\frac{1}{\sqrt{2^n}}$.

In this case, $U = \frac{1}{\sqrt{2^n}} M$, and we can verify that $U^* U = I$.

3 Projectors and Reflections

(a)

$d = 2$

$$\begin{aligned}
 Q &= |\phi\rangle\langle\psi| \\
 \text{when } |\psi\rangle &= |0\rangle = (1 \ 0)^T \\
 \text{and } |\phi\rangle &= |+\rangle = \frac{1}{\sqrt{2}}(1 \ 1)^T \\
 Q &= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \\
 \text{when } |\psi\rangle &= |+\rangle = \frac{1}{\sqrt{2}}(1 \ 1)^T \\
 \text{and } |\phi\rangle &= |0\rangle = (1 \ 0)^T \\
 Q &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{bmatrix}
 \end{aligned}$$

(b)

(c)

$|\phi\rangle$

0

(d)

The transformation P will project any vector to the one-dimensional subspace spanned by $|\phi\rangle$ -- remain the component of the vector in the direction of $|\phi\rangle$ and remove the component of the vector that is orthogonal to $|\phi\rangle$.

(e)

This transformation is a reflection transformation that reflects any vectors across the hyperplane perpendicular to $|\phi\rangle$.

Because I is a hermitian matrix and $(|\phi\rangle\langle\phi|)^* = (|\phi\rangle\langle\phi|)$

,so P is also a hermitian matrix.

And $R^* = (I - 2P)^* = I - 2P = R$, So $I - 2P$ is a hermitian matrix as well.

we want to prove that $(I - 2P)^*(I - 2P) = I$.

We set $R = I - 2P$, so we need to prove that $R^*R = I$, i.e. $R^2 = I$.

$$\begin{aligned}
 R^2 &= (I - 2P)^2 = I - 4P + 4P^2 \\
 &= I - 4P + 4P = I
 \end{aligned}$$

Thus, we can conclude that R is a unitary transformation.

(f)

(i)

if we can prove that for any basis vector $|\psi_i\rangle$, $U|\psi_i\rangle = |\phi_i\rangle$, then we can conclude that U is a matrix that maps the basis $\{|\psi_i\rangle\}$ to the basis $\{|\phi_i\rangle\}$.

$$\begin{aligned}
 U|\psi_i\rangle &= \sum_{j=1}^d |\phi_j\rangle \langle \psi_j | \psi_i \rangle \\
 &= \sum_{j=1}^d |\phi_j\rangle \delta_{ji} = |\phi_i\rangle
 \end{aligned}$$

So U can be written as $U = \sum_{i=1}^d |\phi_i\rangle \langle \psi_i|$.

Then we want to prove that U is a unitary transformation, i.e. $U^*U = I$.

$$U^*U = \sum_{i=1}^d |\psi_i\rangle \langle \phi_i|.$$

$$\begin{aligned}
 U^*U &= \sum_{i=1}^d \sum_{j=1}^d |\psi_i\rangle \langle \phi_i | \phi_j \rangle \langle \psi_j| \\
 &= \sum_{i=1}^d \sum_{j=1}^d |\psi_i\rangle \delta_{ij} \langle \psi_j| \\
 &= \sum_{i=1}^d |\psi_i\rangle \langle \psi_i| = I
 \end{aligned}$$

Therefore, we can conclude that U is a unitary transformation.

(ii)

$$I = \sum_{i=1}^d |\phi_i\rangle \langle \phi_i|$$

It reflect any basis vector $|\psi_i\rangle$ to itself.

4 Quantum Anti-Zeno Effect

set a integer n , and we set the $\epsilon = \frac{\pi}{2*n}$. Then every time, we perform a measurement on the state based on the basis $\{|\psi_k\rangle, |\psi_k^\perp\rangle\}$, where $|\psi_k\rangle = \cos(k\epsilon)|0\rangle + \sin(k\epsilon)|1\rangle$ and $|\psi_k^\perp\rangle = -\sin(k\epsilon)|0\rangle + \cos(k\epsilon)|1\rangle$.

Every time we perform a measurement, we expect to get the state $|\psi_k\rangle$.

For the first measurement, the $k = 1$, then $k = 2$ for the second measurement, and so on. After the n -th measurement, $k = n$. Then we can get this qubit's state will be $|\psi_n\rangle = \cos(n\epsilon)|0\rangle + \sin(n\epsilon)|1\rangle = 1$.

When n is large enough, we can get that ϵ is small enough. And the ability of success in every process is $\cos^2(\epsilon)$. When n is large enough, $\cos^2(\epsilon) \approx 1 - \epsilon^2/2$ and it's close to 1. So the probability of success in every process is close to 1, and the total probability of success is close to 1 as well.

5 Composite Quantum Systems

(a)

we know that

$$\begin{aligned}
 I &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
 H &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \\
 I \otimes H &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \\
 H \otimes I &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \\
 H \otimes H &= \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}
 \end{aligned}$$

(b)

$$\begin{aligned}
 (I \otimes H)|0, 1\rangle &= (I|0\rangle) \otimes (H|1\rangle) = |0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\
 &= \frac{1}{\sqrt{2}}(|0, 0\rangle - |0, 1\rangle) \\
 (H \otimes I)|0, 1\rangle &= (H|0\rangle) \otimes (I|1\rangle) \\
 &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle \\
 &= \frac{1}{\sqrt{2}}(|0, 1\rangle + |1, 1\rangle)
 \end{aligned}$$

(c)

We now set two qubits in the basis for $\mathbb{C}^{de}$

$$\begin{aligned}
 |\phi_{ij}\rangle &= |u_i\rangle \otimes |v_j\rangle \\
 |\phi_{kl}\rangle &= |u_k\rangle \otimes |v_l\rangle \\
 \text{Their inner product} &= \langle \phi_{ij} | \phi_{kl} \rangle \\
 &= (\langle u_i | \otimes \langle v_j |)(|u_k\rangle \otimes |v_l\rangle) \\
 &= \langle u_i | u_k \rangle \cdot \langle v_j | v_l \rangle \\
 &= \delta_{ik} \cdot \delta_{jl}
 \end{aligned}$$

So if and only if $i = k$ and $j = l$, the inner product is 1, otherwise, the inner product is 0. So this set is a orthonormal sets. And we know that the number of vectors in this set is de , which is the same as the dimension of $\mathbb{C}^{de}$, so this set is an orthonormal basis for $\mathbb{C}^{de}$.

(d)

For every element in matrix $A \otimes B$, we can write it as

$$\begin{aligned}(A \otimes B)_{ij,kl} &= (\langle u_i | \otimes \langle v_j |)(A \otimes B)(|u_k\rangle \otimes |v_l\rangle) \\ &= \langle u_i | A | u_k \rangle \cdot \langle v_j | B | v_l \rangle = A_{ik} \cdot B_{jl}\end{aligned}$$

We know that

$$A = \sum_{i,k} A_{ik} |u_i\rangle \langle u_k|, B = \sum_{j,l} B_{jl} |v_j\rangle \langle v_l|$$

So we can get that

$$A \otimes B = \sum_{i,k} \sum_{j,l} \langle u_i | A | u_k \rangle \cdot \langle v_j | B | v_l \rangle \cdot (|u_i\rangle \langle u_k|) \otimes (|v_j\rangle \langle v_l|)$$

(e)

No.

$$|\psi\rangle = \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right)^{\otimes n} = (|+\rangle)^{\otimes n}$$

We can write the Alice part as $|\phi_A\rangle = (|+\rangle)^{\otimes j}$

, and the Bob part as $|\phi_B\rangle = (|+\rangle)^{\otimes n-j}$.

Then we can get that $|\psi\rangle = |\phi_A\rangle \otimes |\phi_B\rangle$.

By the definition of entangled state, we can conclude that $|\psi\rangle$ is not an entangled state, because it can be written as a tensor product of two states.